

Consistency Checks for PBH Accretion in the Zero Relative-Velocity Limit

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Purpose

The purpose of this calculation is to redo the consistency checks in Ali-Haïmoud and Kamionkowski (2017) in one particular limit:

$$v_{\text{rel}} = 0.$$

I refer to the paper as YAH throughout this note [1].

In the original calculation, the relative motion between a PBH and the surrounding gas is accounted for using an effective velocity or, in the luminosity calculation, by averaging over a distribution of relative velocities. Here I remove that extra bulk motion completely and ask what changes when the PBH is at rest relative to the gas.

This does not mean that the gas itself is motionless. The gas still has thermal motion and still falls inward under the gravity of the PBH. The only thing being removed is the additional streaming motion between the PBH and the background gas. Since the accretion rate depends very strongly on the velocity entering the Bondi calculation, setting $v_{\text{rel}} = 0$ generally makes the accretion more efficient.

My main goal was to see what happens next. If the accretion rate and luminosity become larger, do the assumptions used to derive the original solution still remain reasonable? I therefore repeated the checks on the size of the Bondi region, the steady-flow approximation, radiative cooling, the escape of radiation, and local thermal feedback using the zero-relative-velocity outputs.

I also compare the new results with the original YAH outputs so that it is clear which changes come specifically from removing the relative velocity. The calculation covers PBH masses of 1, 100, and $10^4 M_{\odot}$ over $50 \leq z \leq 10^4$.

What is changed, and what is kept fixed?

The main change in this calculation is the velocity entering the Bondi accretion model. YAH define the thermal Bondi velocity, radius, and time as

$$v_B^2 = \frac{P_{\infty}}{\rho_{\infty}} = \frac{(1 + x_e)k_B T_{\text{gas}}}{m_p}, \quad r_B = \frac{GM}{v_B^2}, \quad t_B = \frac{GM}{v_B^3}. \quad (1)$$

Here v_B comes from the ordinary thermal motion of the gas. The Bondi radius r_B gives the characteristic region over which the PBH can gravitationally capture the gas, while $t_B = r_B/v_B$ gives the corresponding flow time. I use the same definition of v_B as YAH throughout.

A PBH can also have a bulk velocity relative to the surrounding gas. YAH approximately account for this by replacing

$$v_B^2 \longrightarrow v_B^2 + v_{\text{rel}}^2.$$

In the calculation here I instead set

$$v_{\text{rel}} = 0, \quad T_{\text{eff}} = T_{\text{gas}}, \quad (2)$$

so the velocity entering the Bondi calculation reduces to the ordinary thermal Bondi velocity v_B .

This does not mean that the gas itself is motionless or that its temperature is zero. It only means that the PBH has no additional bulk motion relative to the gas.

The important point is that this change affects more than just the final luminosity. The Bondi time depends strongly on velocity,

$$t_B \propto v_B^{-3},$$

so changing the velocity also changes the importance of Compton drag and cooling. YAH describe these effects using

$$\beta = \frac{4x_e \sigma_T u_{\text{CMB}}}{3m_p c} t_B, \quad \gamma = \frac{8x_e \sigma_T u_{\text{CMB}}}{3m_e c(1+x_e)} t_B. \quad (3)$$

Here u_{CMB} is the CMB energy density.

The accretion rate is then

$$\dot{M} = 4\pi\lambda(\beta, \gamma)\rho_b \frac{(GM)^2}{v_B^3}, \quad \dot{m} = \frac{\dot{M}c^2}{L_{\text{Edd}}}. \quad (4)$$

Here λ describes how Compton drag, Compton cooling, and gas pressure modify the Bondi flow. Since β and γ change when the velocity changes, λ changes as well. The zero-velocity result cannot be obtained by simply multiplying the original accretion rate by one fixed velocity factor.

The new accretion rate then changes the inner temperature and luminosity, which is why I recalculate the consistency checks rather than only rescaling the published results.

Finally, $\dot{m}$ should not be confused with luminosity. It compares the rest-mass energy flowing into the PBH with the Eddington luminosity. The fraction of that energy actually radiated is determined separately by the radiative efficiency.

How much do accretion and luminosity change?

The first thing I wanted to see was how strongly setting $v_{\text{rel}} = 0$ changes the accretion rate.

Figure 1 compares the original characteristic-velocity result with the new zero-velocity result using the same background gas. Removing the PBH–gas relative motion increases the accretion rate in every case shown. The size of the increase depends on both PBH mass and redshift because β , γ , and λ also change when the velocity changes.

$M/M_\odot$	largest $\dot{m}$ ratio	z	largest L ratio	z
1	34.1	1006	579	996
100	30.8	808	398	779
10^4	13.3	403	115	704

Table 1: Largest enhancements over the sampled redshift range. The luminosity ratio is $L(0)/\langle L \rangle_{50}$ in the collisional-ionization case, where the denominator is the original luminosity averaged over 50 relative velocities. The accretion and luminosity ratios have different denominators.

The luminosity changes even more strongly than the accretion rate. YAH Eq. (57) gives an approximately quadratic dependence on the accretion rate, together with an additional dependence on the temperature near the black hole. This is why an accretion-rate increase of tens can produce a luminosity increase of hundreds.

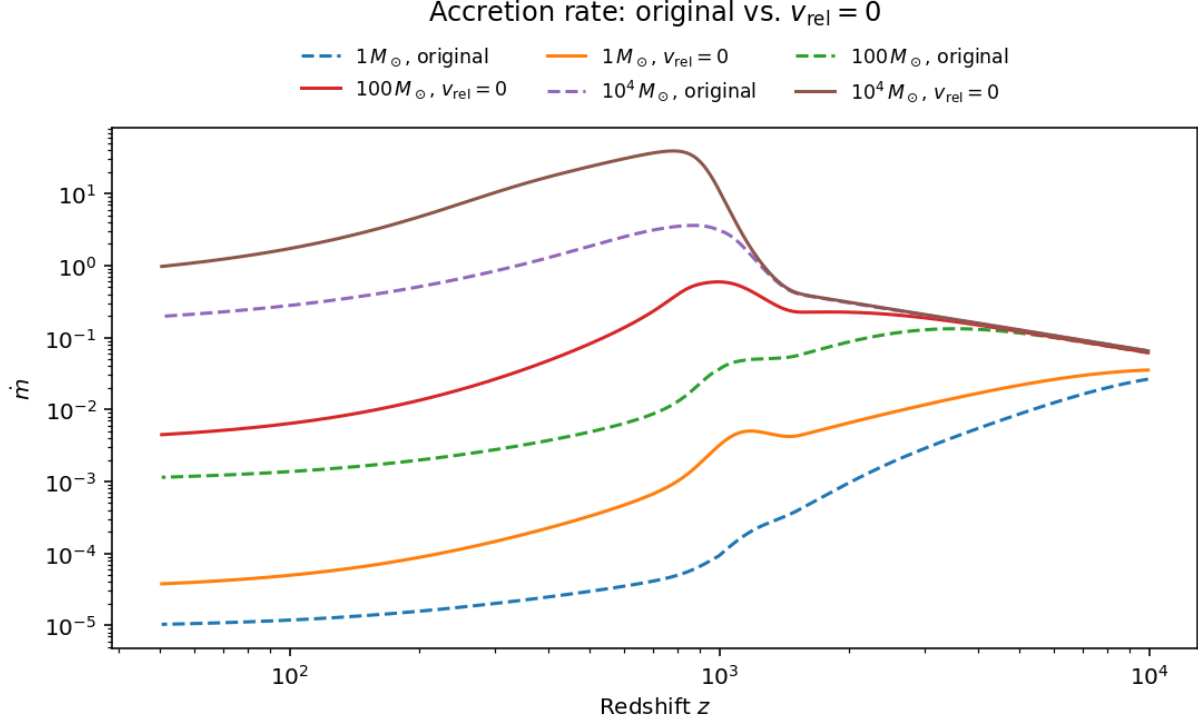


Figure 1: Accretion rates using the original characteristic v_{eff} prescription (dashed) and $v_{\text{rel}} = 0$ (solid). Both calculations use the same background gas. The original curves use a characteristic v_{eff} rather than a direct velocity average of $\dot{M}$.

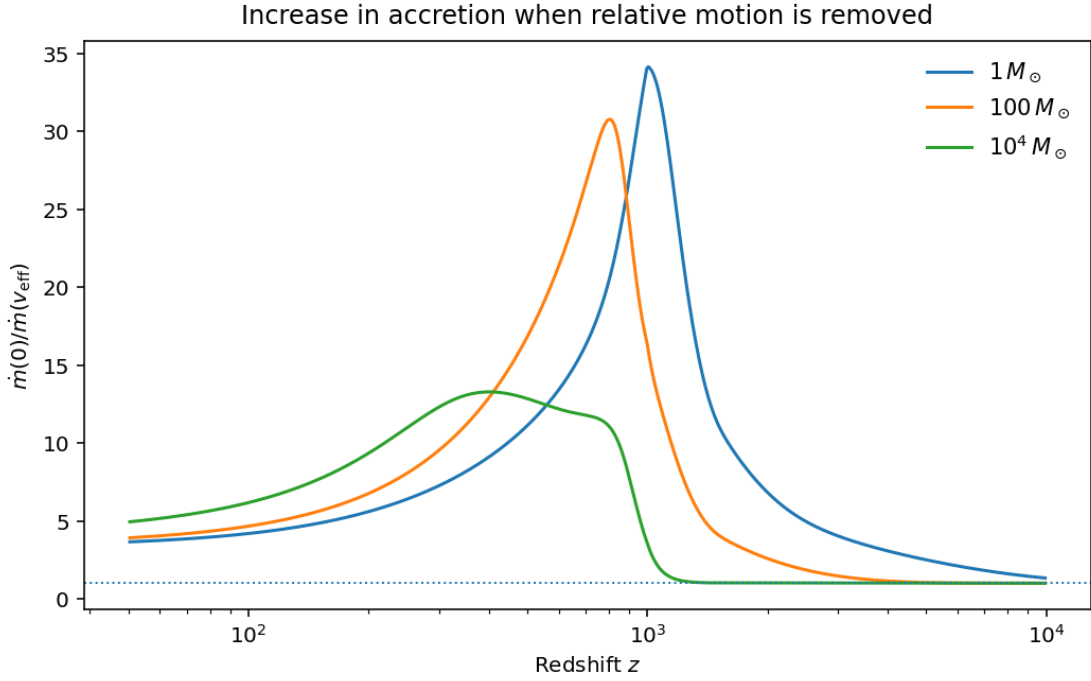


Figure 2: Ratio of the zero-velocity and original accretion rates. A value of 30 means that setting $v_{\text{rel}} = 0$ gives an accretion rate thirty times larger than the characteristic- v_{eff} result at that redshift.

It would not be correct to simply square the accretion-rate ratios in Table 1. The inner temperature also changes, and the original luminosity is averaged over the relative-velocity distribution rather than evaluated at the single characteristic velocity used in the accretion-rate plot.

I also repeated the luminosity average with 100, 200, and 400 velocity samples to check that the 50-sample result was stable. The largest 50-to-400 difference was 2.87%, and the 200-to-400 difference was at most 0.0416%.

What am I checking?

YAH use several consistency checks to make sure that the assumptions behind their accretion model remain reasonable. Since these checks test different parts of the model, I found it helpful to first summarize what each one is asking.

Check	Location in YAH	What is being tested?
Isolated PBH	Appendix A.1	Is the Bondi region much smaller than the typical distance between PBHs?
Steady flow	Section II.B.1	Can the gas flow adjust before the cosmological background changes significantly?
Mass growth	Section II.B	Does the PBH gain only a small fraction of its mass over a Hubble time?
Free-free cooling	Appendix A.2	Can bremsstrahlung cooling be neglected while the gas falls inward?
Free-bound emission	Appendix A.3	Is recombination radiation small compared with free-free radiation?
Scattering	Appendix A.4	Is the electron column small enough that scattering can still be neglected?
Free-free absorption	Appendix A.4	Is absorption of the emitted radiation still small?
Local thermal feedback	Section III.A, Figure 9	Does radiation from the PBH significantly heat the same gas that is feeding it?

Table 2: Summary of the consistency checks used in the zero-relative-velocity calculation.

Isolation, flow time, and mass growth

Isolated PBHs: YAH Appendix A.1

The first check asks whether each PBH can still be treated as accreting from its own separate region of gas. The relevant comparison is between the Bondi radius r_B and the typical distance between PBHs.

Following YAH Eq. (A2), the characteristic PBH separation is

$$\bar{r}_{\text{PBH}} = \left(\frac{3M}{4\pi f_{\text{PBH}} \rho_{\text{dm}}} \right)^{1/3}, \quad \mathcal{C}_{\text{iso}} = \frac{r_B}{\bar{r}_{\text{PBH}}}. \quad (5)$$

Here f_{PBH} is the fraction of dark matter made of PBHs. If $\mathcal{C}_{\text{iso}} \ll 1$, the Bondi region is much smaller than the distance to the next PBH, so the isolated-PBH approximation is reasonable. If the ratio approaches or exceeds one, the capture regions are no longer clearly separated.

For the main comparison I set $f_{\text{PBH}} = 1$. For $10^4 M_\odot$, the largest ratio is 1.24, reached near the low-redshift end of the grid. The result depends on the assumed PBH abundance because

$$\mathcal{C}_{\text{iso}} \propto f_{\text{PBH}}^{1/3}. \quad (6)$$

For $f_{\text{PBH}} = 10^{-4}$, the maximum instead falls to 0.0576. This shows how strongly the isolation estimate depends on the assumed PBH abundance, so the $f_{\text{PBH}} = 1$ warning should not be applied directly to much rarer PBH populations.

It is also important that YAH's original isolation estimate already assumes a PBH at rest. Their estimate additionally uses $T_{\text{gas}} \simeq T_{\text{CMB}}$ and applies roughly over $200 \lesssim z \lesssim 1100$. My crossing near $z \sim 66$ occurs outside that range, so it does not directly contradict the estimate made in Appendix A.1. As in YAH, this comparison also assumes a roughly uniform PBH distribution and does not include PBH clustering.

Steady flow: YAH Section II.B.1

The next check asks whether the gas flow can settle into a steady state before the cosmological background changes significantly. The relevant comparison is between the Bondi time t_B and the Hubble time H^{-1} :

$$\mathcal{C}_{\text{ss}} = Ht_B = H \frac{GM}{v_B^3}. \quad (7)$$

If $Ht_B \ll 1$, the gas has enough time to adjust to the PBH before the background changes very much, so the steady-flow approximation is reasonable. If the ratio approaches one, the cosmological background is changing on a similar timescale to the accretion flow.

For $10^4 M_\odot$, the maximum value is 2.14, while for $100 M_\odot$ it is only about 2.14×10^{-2} . The high-mass zero-velocity case reaches a regime at low redshift where the Bondi time becomes comparable to the Hubble time.

This does not prove that the flow becomes unstable. It only shows that the timescale condition used to justify a steady-state treatment is no longer strongly satisfied.

Mass growth: YAH Eq. (12)

I also check whether the PBH gains enough mass during one Hubble time that treating its mass as fixed becomes questionable. I use

$$\mathcal{C}_M = \frac{\dot{M}}{HM}. \quad (8)$$

This estimates the fraction of the PBH's mass that would be accreted over one Hubble time if the instantaneous accretion rate stayed the same.

For $10^4 M_\odot$, the maximum value is 0.151. This is still below one, but it is large enough that the change in mass is no longer completely negligible.

The maximum also occurs in a region where the steady-flow approximation is already becoming questionable. I treat this mainly as a diagnostic of the zero-velocity solution, rather than as a precise prediction of how much mass the PBH would actually gain.

Cooling and radiation checks

Free-free cooling: YAH Appendix A.2

Free-free radiation, or bremsstrahlung, is the main source of radiation in the inner YAH accretion flow. The question here is whether the gas loses energy through this radiation quickly enough to change the temperature profile used in the accretion calculation.

YAH compare the accretion time with the free-free cooling time:

$$\frac{t_{\text{acc}}}{t_{\text{ff}}} \sim \dot{m} \alpha \mathcal{J}(X), \quad X = \frac{k_B T}{m_e c^2}. \quad (9)$$

Here $\mathcal{J}$ is the free-free emissivity fit from YAH Eq. (55). For my numerical comparison, I evaluate the expression using the supplied temperature near the Schwarzschild radius, T_S :

$$\mathcal{C}_{\text{ff},S} = \dot{m} \alpha \mathcal{J}\left(\frac{k_B T_S}{m_e c^2}\right). \quad (10)$$

If $\mathcal{C}_{\text{ff},S} \ll 1$, the gas falls inward faster than free-free radiation can cool it, so neglecting this cooling is reasonable. If the ratio approaches one, cooling is no longer negligible.

For the $10^4 M_\odot$ collisional branch, the maximum reaches 1.47. This means that free-free cooling is no longer clearly negligible in the high-mass zero-velocity case.

This is still only an order-of-magnitude check. I evaluate it at T_S rather than calculating the cooling rate at every radius, so the result should be read as a near-horizon diagnostic rather than an exact boundary where the flow suddenly changes.

Free-bound emission: YAH Appendix A.3

The next check asks whether free-bound radiation becomes important compared with the free-free radiation used in the YAH luminosity calculation. Free-bound radiation is produced when a free electron is captured by an ion.

YAH Eq. (A8) estimates the ratio of free-bound to free-free emission. Using the supplied temperature near the Schwarzschild radius, T_S , I calculate

$$\mathcal{C}_{\text{fb},S} = \left(\frac{E_I}{\sqrt{m_e c^2 k_B T_S}} \right)^3 \frac{\sigma_0}{\alpha \sigma_T}, \quad E_I = 13.6 \text{ eV}, \quad \sigma_0 \simeq 6 \times 10^{-18} \text{ cm}^2. \quad (11)$$

For the $10^4 M_\odot$ collisional case, the maximum value is 0.0246. Since this remains below one, free-bound emission stays smaller than free-free emission at the horizon temperatures tested here.

As with the cooling check, this is a local estimate using T_S . It does not calculate the total emission over every radius and photon frequency.

Scattering: YAH Appendix A.4

The next check asks whether photons can escape from the inner flow without being strongly scattered by free electrons.

YAH Eq. (A9) uses the Thomson cross-section to estimate the electron column near the black hole. Using the near-horizon density from YAH Eq. (56), I calculate

$$n_{e,S} = \frac{\dot{M}}{4\pi m_p r_S^2 c}, \quad r_S = \frac{2GM}{c^2}, \quad \mathcal{C}_{\text{T},S} = r_S n_{e,S} \sigma_T. \quad (12)$$

With the definitions used in this calculation, this becomes

$$\mathcal{C}_{\text{T,S}} = \frac{\dot{m}}{2}. \quad (13)$$

YAH quote this only to order of magnitude as $\sim \dot{m}$ in Eq. (A9); the factor of two here comes from keeping the explicit $r_S = 2GM/c^2$ normalization.

I treat $\mathcal{C}_{\text{T,S}}$ as a Thomson-column estimate rather than the exact Compton optical depth. The true scattering probability depends on the photon energy, so this quantity is mainly useful as a consistency check: if it is much smaller than one, scattering is safely small.

The maximum value is about 0.298 for $100 M_\odot$ and 19.6 for $10^4 M_\odot$. In the high-mass zero-velocity case, this estimate is no longer small, so this check by itself is no longer enough to justify neglecting scattering.

It does not, however, prove that the true Compton optical depth is greater than one. Determining that would require a more complete calculation of how photons of different energies travel through the flow.

Free-free absorption and degeneracy: YAH Eqs. (A10)–(A12)

The final check in Appendix A asks whether the free-free radiation produced near the black hole can escape without being strongly reabsorbed by the gas.

YAH estimate the absorption near the characteristic photon energy $h\nu \sim k_B T_S$. Using their Eq. (A12), I calculate

$$\mathcal{D}_S = \left(\frac{hc n_{e,S}^{1/3}}{k_B T_S} \right)^3, \quad \mathcal{C}_{\text{abs,S}} \sim \alpha \mathcal{C}_{\text{T,S}} \mathcal{D}_S. \quad (14)$$

The quantity $\mathcal{D}_S$ is the degeneracy factor used in YAH, while $\mathcal{C}_{\text{abs,S}}$ estimates the corresponding free-free absorption. If both are much smaller than one, these effects remain small in the inner flow.

For the $10^4 M_\odot$ collisional branch, the maximum values are 2.3×10^{-8} and 1.07×10^{-9} , respectively. Both remain far below one even in the zero-velocity case.

As with the previous tests, these are near-horizon estimates. They do not show that the flow is transparent at every photon frequency or at every radius.

Putting the checks together

Table 3 collects the largest values of the consistency checks over the redshift range used here. The temperature-dependent quantities use the collisional-ionization branch, and the isolation check uses $f_{\text{PBH}} = 1$.

Figure 3 shows the four checks that become most important for the $10^4 M_\odot$ zero-velocity case.

The largest effects appear in the $10^4 M_\odot$ case. The isolation and steady-flow checks become large at low redshift, while the cooling and scattering estimates become important over different redshift ranges. So there is not one redshift where every assumption suddenly stops working.

The lower-mass cases are much less extreme. For $100 M_\odot$, the largest of these ratios is the Thomson-column estimate, which reaches about 0.3 but remains below one. The $1 M_\odot$ case stays well below one in the main checks shown here.

Diagnostic	$1 M_\odot$	$100 M_\odot$	$10^4 M_\odot$
$r_B/\bar{r}_{\text{PBH}}$	2.67×10^{-3}	0.0576	1.24
Ht_B	2.14×10^{-4}	0.0214	2.14
$\dot{M}/(HM)$	6.4×10^{-6}	1.05×10^{-3}	0.151
$\mathcal{C}_{\text{ff},S}$	2.29×10^{-3}	0.0171	1.47
$\mathcal{C}_{\text{fb},S}$	7.81×10^{-5}	3.43×10^{-4}	0.0246
$\mathcal{C}_{\text{T},S}$	0.0177	0.298	19.6
$\mathcal{D}_S$	7.26×10^{-13}	4.54×10^{-11}	2.3×10^{-8}
$\mathcal{C}_{\text{abs},S}$	3.45×10^{-17}	9.66×10^{-14}	1.07×10^{-9}

Table 3: Maximum values of the consistency checks over $50 \leq z \leq 10^4$. Different checks can reach their largest values at different redshifts.

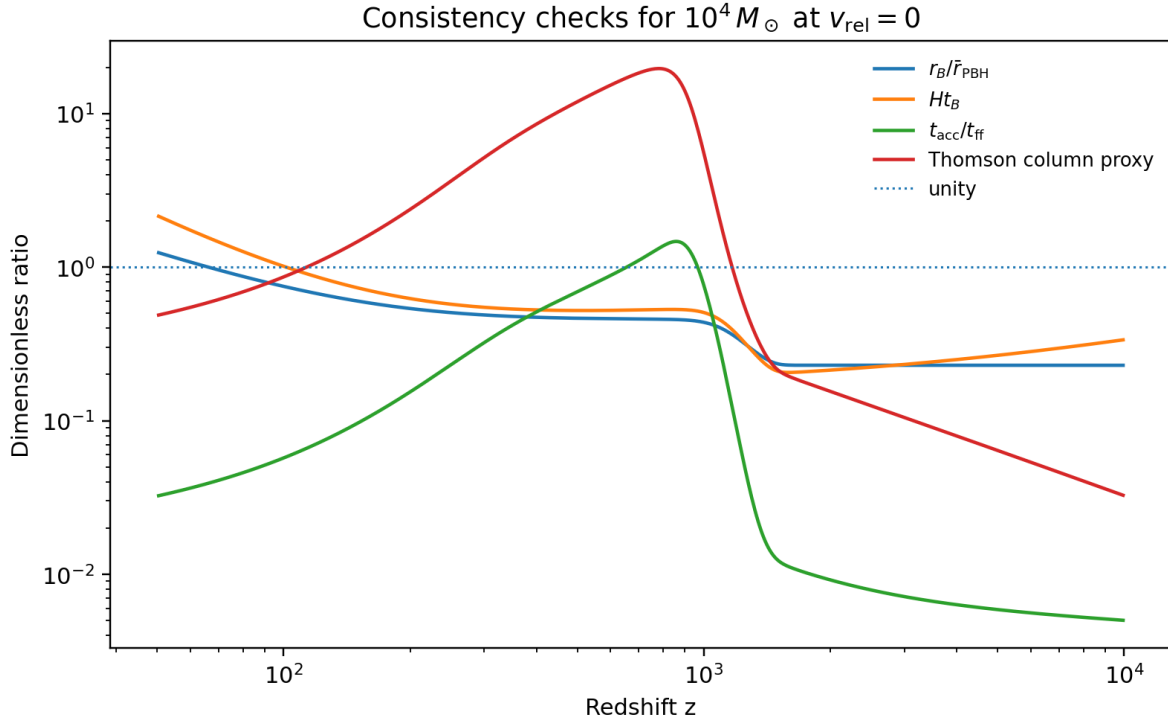


Figure 3: Main consistency checks for $10^4 M_\odot$ at $v_{\text{rel}} = 0$. When one of these quantities approaches or exceeds one, the corresponding ratio is no longer safely small. The Thomson curve is only an estimate of scattering and is not the exact Compton optical depth.

Diagnostic	sampled z interval with value ≥ 1
$r_B/\bar{r}_{\text{PBH}}$	50–66
Ht_B	50–101
$\mathcal{C}_{\text{ff},S}$	661–961
$\mathcal{C}_{\text{T},S}$	114–1164

Table 4: Sampled redshift ranges where the four quantities in Figure 3 reach or exceed one. The endpoints are approximate grid values rather than exact transition redshifts. An endpoint at $z = 50$ is the edge of the sampled range, not a measured lower boundary.

So the main result is that setting $v_{\text{rel}} = 0$ pushes the high-mass solution much closer to the limits of several assumptions in the model. Which assumption becomes important depends on the redshift and on the physical effect being tested.

Luminosity and the HyRec response

Figure 4 shows how much the luminosity increases when $v_{\text{rel}} = 0$, while keeping the same background gas used in the original accretion calculation.

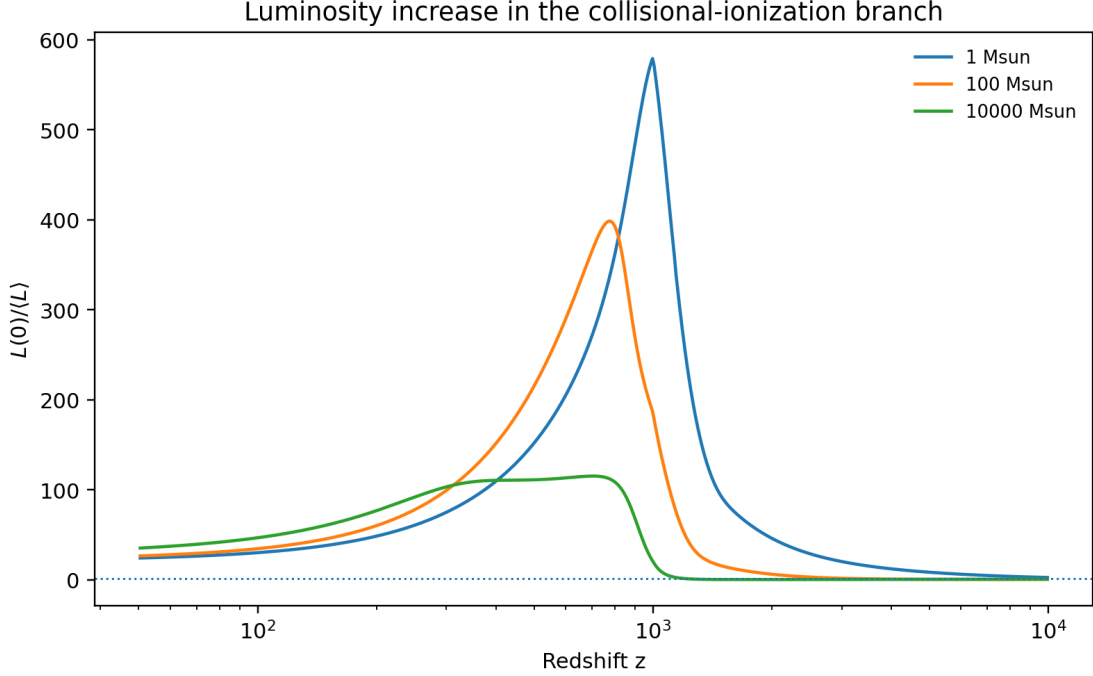


Figure 4: Luminosity enhancement in the collisional-ionization branch. The denominator is the original luminosity averaged over the relative-velocity distribution. Large ratios show how strongly the luminosity changes when the relative motion is removed.

I then used the modified luminosity in the HyRec energy-injection calculation. This is a different step from the fixed-background accretion calculation above. Here the extra PBH energy is allowed to change the globally averaged background gas temperature and free-electron fraction.

Figure 5 shows this response for the $100 M_{\odot}$, $f_{\text{PBH}} = 1$ example.

$M/M_{\odot}$	f_{PBH}	largest $ \Delta x_e $	z
10	1	1.43×10^{-3}	1196
100	1	0.0141	1153
10^3	0.01	6.16×10^{-3}	426
10^4	10^{-4}	3.73×10^{-3}	301

Table 5: Largest differences between the zero-velocity and original ionization histories over $50 \leq z \leq 2000$. Here Δx_e is an absolute change in the electron fraction, not a percentage.

The larger PBH luminosity produces a larger change in the ionization history. The no-PBH curve stays the same, as expected.

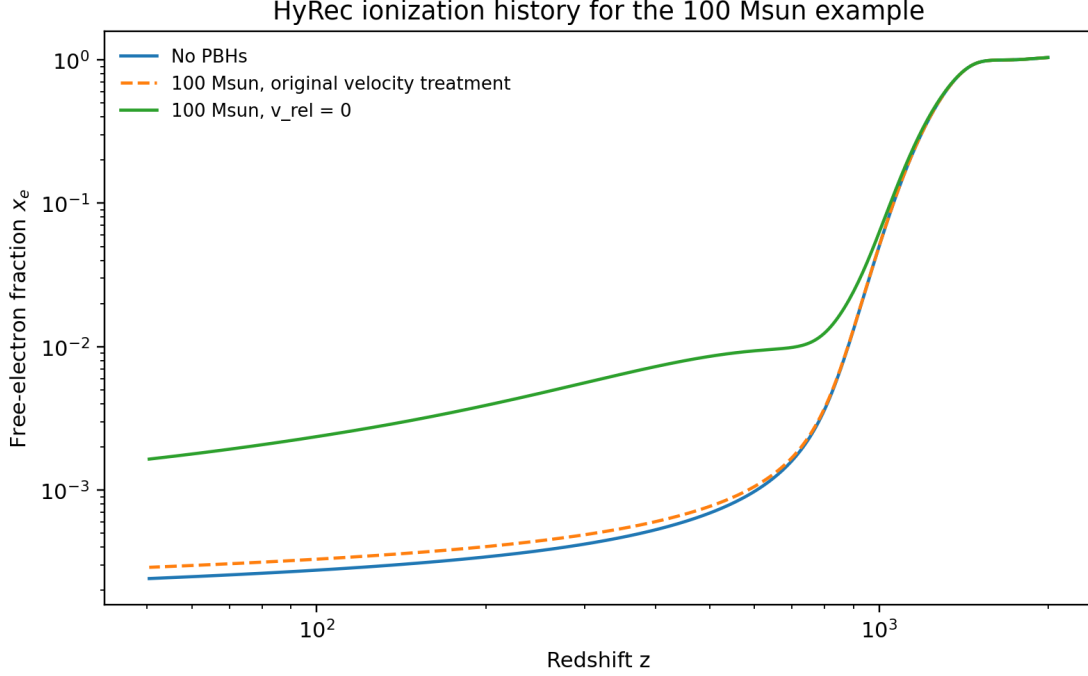


Figure 5: Free-electron fraction for the $100 M_{\odot}$, $f_{\text{PBH}} = 1$ example. The zero-velocity result is compared with the original velocity treatment and with the no-PBH case.

I have not used these new histories to calculate CMB power spectra or new PBH limits. They only show how the existing HyRec/PBH calculation responds when the zero-velocity luminosity is used.

Local thermal feedback and Figure 9

The last check asks whether radiation from the PBH heats the same gas that is feeding the accretion flow. This matters because the temperature of that gas is one of the quantities used to calculate the accretion rate in the first place.

YAH compare the local Compton heating produced by the PBH with whichever is larger of CMB Compton cooling and adiabatic heating, and then find the radius where this ratio is maximized. If the resulting ratio is much smaller than one, the extra heating can be neglected. If it approaches or exceeds one, the assumed temperature profile is no longer clearly independent of the PBH radiation.

Writing the temperatures in kelvin, YAH Section III.A gives the following approximation for this maximum in their printed Eq. (66):

$$\mathcal{F}_{\text{fb}}^{(66)} \approx 0.07 \frac{x_e}{1 + x_e} \frac{L}{L_{\text{Edd}}} \frac{v_B}{c} \frac{m_p c^2}{k_B T_{\text{CMB}}} \sqrt{1 + \gamma^{2/3}}. \quad (15)$$

This is also where my earlier Figure 9 work becomes relevant. When I reconstructed the original 2017 calculation, I found that the numerical source does not evaluate Eq. (66) literally. Instead, the feedback output is calculated as

$$\mathcal{F}_{\text{fb}}^{(C)} = \frac{L}{L_{\text{Edd}}} P_C (1 + \gamma^{1/3}), \quad (16)$$

where, in the $v_{\text{rel}} = 0$ limit,

$$P_C = 0.067 \frac{x_e}{1+x_e} \frac{1}{3 \times 10^{10}} \sqrt{\frac{T_{\text{gas}}}{1.21 \times 10^{-8}}} \frac{1.1 \times 10^{13}}{T_{\text{gas}}}. \quad (17)$$

The numerical constants in this expression use the cgs and kelvin normalizations of the original source.

The two expressions are not the same. The printed equation contains $\sqrt{1 + \gamma^{2/3}}$, while the source uses $1 + \gamma^{1/3}$. The prefactor is also different: the source uses 0.067 rather than 0.07 and is written in terms of T_{eff} , which reduces to T_{gas} when $v_{\text{rel}} = 0$, rather than T_{CMB} . When the source prefactor is rewritten in terms of its Bondi velocity, it also contains an extra factor $(1 + x_e)^{-1/2}$. I have not found a derivation in the source explaining this difference.

For that reason, I do not choose one expression and assume that it is the intended one. I evaluate both using the same zero-velocity luminosity, x_e , v_B , and γ . This limit is especially useful because there is no velocity distribution left to average over, so any difference between the two curves cannot come from velocity averaging.

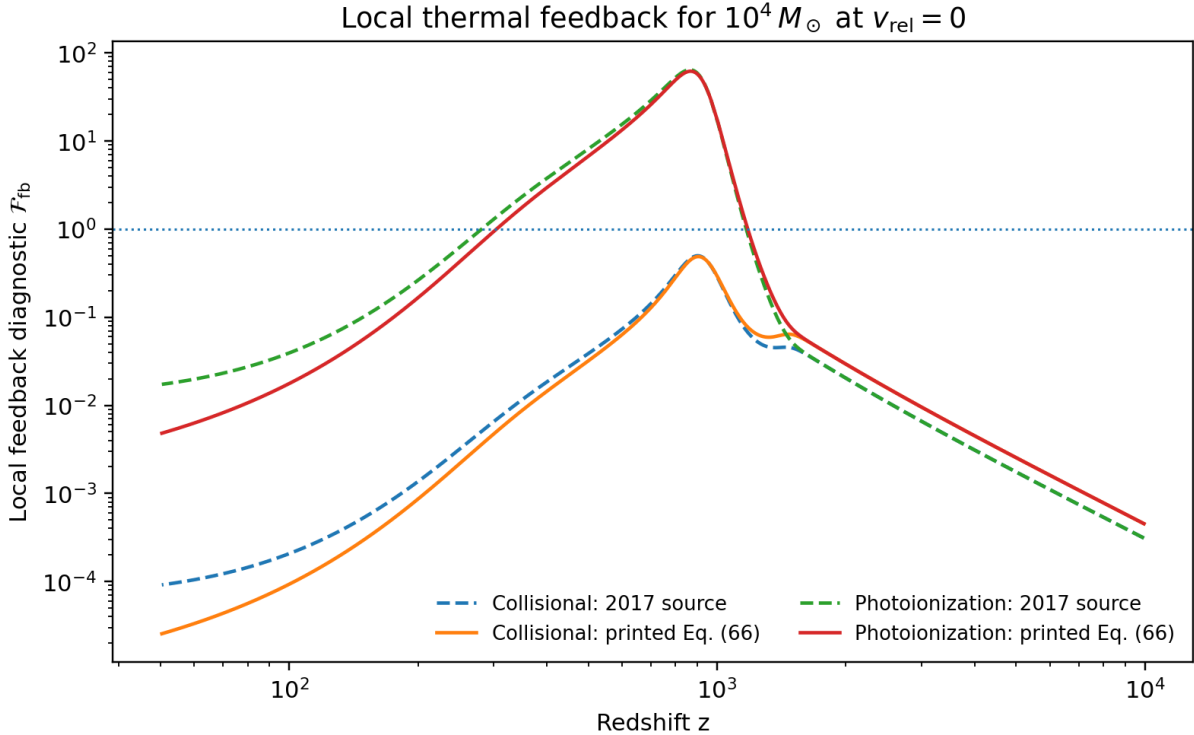


Figure 6: Local thermal-feedback estimates for $10^4 M_{\odot}$ at $v_{\text{rel}} = 0$. Dashed curves reproduce the original 2017 source prescription, while solid curves use the printed Eq. (66). Both are shown for the collisional and photoionization limits. These are fixed-background feedback diagnostics, not self-consistent radiation-hydrodynamic solutions.

For the $10^4 M_{\odot}$ collisional branch, both estimates peak near 0.5. This is still below one, but it is large enough that I would not call local heating negligible.

The photoionization branch is much more extreme. The two estimates reach about 64 and 62. At those values, local PBH heating is much larger than the thermal terms used in the original solution, so the assumed temperature profile is not self-consistent under this local-heating check.

The printed equation and the source implementation are numerically different, but they give the

$M/M_\odot$	branch	source prescription	printed Eq. (66)
1	Collisional	7.39×10^{-4}	9.87×10^{-4}
1	Photoionization	7.39×10^{-4}	9.87×10^{-4}
100	Collisional	0.0644	0.0873
100	Photoionization	0.302	0.305
10^4	Collisional	0.498	0.487
10^4	Photoionization	64.2	61.9

Table 6: Maximum local-feedback estimates over the sampled redshift range for the two prescriptions and the two ionization limits.

same main warning in the high-feedback case. I keep both results rather than assuming which version was intended.

What the zero-velocity test shows

Setting $v_{\text{rel}} = 0$ increases the accretion rate a lot, especially for $10^4 M_\odot$. But that is also the case where several of the assumptions behind the calculation start to become questionable. The isolation and steady-flow ratios grow at low redshift, free-free cooling is no longer clearly small, the Thomson-column estimate becomes large, and the photoionization branch gives very strong local thermal feedback.

These effects do not all become important at the same redshift, so I do not think there is one point where the whole model suddenly stops being valid. Instead, the different checks show where different parts of the approximation start to become less reliable.

I also would not treat the $v_{\text{rel}} = 0$ result as a corrected version of the YAH calculation. It is more useful as a limiting case that shows what happens when the suppression from relative motion is removed.

Once some of these assumptions are no longer reliable, this calculation does not tell us what the accretion rate should become instead. It mainly shows where a more complete treatment of the flow, cooling, scattering, or local heating would be needed.

References

- [1] Y. Ali-Haïmoud and M. Kamionkowski, “Cosmic microwave background limits on accreting primordial black holes,” *Physical Review D* **95**, 043534 (2017), doi:10.1103/PhysRevD.95.043534; arXiv:1612.05644. Equation and appendix references in this note follow the supplied 20-page version.